

Differentiable functions – 8

Smoothness and pathologies

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Theoretical minimum.

Theorem 1 (Darboux). A derivative has the intermediate value property; consequently f' cannot have a jump discontinuity, although it may be discontinuous on a dense set.

Theorem 2 (Borel). For every sequence (a_n) of reals there is $f \in C^\infty(\mathbb{R})$ with $f^{(n)}(0) = a_n$ for all n ; hence a formal Taylor series need have nothing to do with the function, and e^{-1/x^2} is the standard flat example.

Fact 1. The three levels of regularity are strictly nested: $C^1 \supsetneq C^\infty \supsetneq \{\text{real-analytic}\}$ in the sense that each inclusion of classes is proper, and the Weierstrass function shows that continuity implies nothing about differentiability.

1. Prove Darboux' theorem, and construct a differentiable function whose derivative is unbounded on every interval and discontinuous on a dense set.
2. Prove Borel's theorem, and use it to construct a C^∞ function which is flat at 0 (all derivatives vanish) but positive elsewhere.
3. (Proposed problems) Let f be a C^1 function on $[0, 1]$, infinitely differentiable at $x = 0$, with $f^{(n)}(0) = 0$ for every n and $|xf'(x)| \leq C|f(x)|$ on $[0, 1]$ for some $C > 0$. Prove that $f \equiv 0$ on $[0, 1]$.
4. (Mexmat, 2006) Let f be infinitely differentiable on $\mathbb{R}^2$ and let U be an open disc with $\iint_{U \times U} \frac{|f(x) - f(y)|}{|x - y|^3} dx dy < \infty$. Prove that f is constant on U .
5. Prove that the Weierstrass function $W(x) = \sum_{n \geq 0} a^n \cos(b^n \pi x)$ with $0 < a < 1$, $ab > 1 + \frac{3\pi}{2}$ and b an odd integer is continuous and nowhere

differentiable; at least carry out the estimate at a point of your choice in full.